Exact calibration tests for strain–pressure identities

Project Stardust

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Abstract

We assemble three exact diagnostics for velocity gradients. A family of nonnormal restricted Euler matrices has fixed full spectrum but variable strain spectrum, with arbitrarily small positive normalized stretching along trajectories whose raw stretching diverges. Its affine realization solves the local Navier–Stokes equations with quadratic pressure, but has infinite whole-space energy. A trace computation then detects an incompatible mixture of full and trace-free pressure conventions: the proposed strain right side has trace equal to the squared Frobenius norm of the velocity gradient under the pressure Poisson constraint. Finally a smooth periodic viscous ABC flow has zero averaged stretching derivative, while a displayed plus-pressure balance gives three at the initial time. Complete elementary proofs reconstruct frozen formal results. These are calibration examples in classical model families, not new fluid existence or singularity results; priority for the diagnostic formulations remains unresolved.

1 Conventions and purpose

For a real 3×3 matrix A, use

$$S = \tfrac{1}{2}(A + A^{\mathsf{T}}), \qquad W = A - S, \qquad \omega = (A_{32} - A_{23}, A_{13} - A_{31}, A_{21} - A_{12}).$$

For an actual velocity u, $A_{ij} = \partial_j u_i$ and $\omega = \nabla \times u$. The pressure Hessian is $P = \nabla^2 p$, so the incompressible pressure constraint is

$$-\operatorname{tr} P = \operatorname{tr}(A^2). \tag{1}$$

This convention agrees with the velocity-gradient formulation in Cantwell [1] and the explicit pressure equations of Yang, Xu, Pumir and He [2]. Full pressure and its deviatoric part $P_0 = P - \frac{1}{3}\operatorname{tr}(P) I$ must be distinguished.

The first test concerns the matrix ODE

$$A' = -A^2 + \tfrac{1}{3}\operatorname{tr}(A^2) I, \qquad \operatorname{tr} A = 0. \tag{2}$$

The second is an algebraic test of a specified strain right side. The third tests a specified spatially averaged identity on an actual smooth periodic Navier–Stokes flow. Each test has its own hypotheses; neither an ODE trajectory nor an infinite-energy affine field is promoted to a finite-energy singularity theorem.

2 Fixed full spectrum and arbitrarily small normalized stretching

For $p > 2$ set $a = \sqrt{(p-2)(p+1)} > 0$ and define

$$M_{p,a} = \begin{pmatrix} -p & 0 & a \\ 0 & 1 & 0 \\ -a & 0 & p-1 \end{pmatrix}, \qquad c(t) = (1-t)^{-1}, \qquad A(t) = c(t) M_{p,a}. \tag{3}$$

For $\omega \neq 0$ let

$$\sigma(A) = \frac{\omega^{\mathsf{T}} S \omega}{|\omega|^2}, \qquad \rho(A) = \frac{\sigma(A)}{\|S\|_F}.$$

The normalization uses the Frobenius norm of the symmetric strain, not the spectral radius of the nonsymmetric full matrix.

Theorem 2.1. *For every $p > 2$ the trajectory (3) solves (2) for $t < 1$. Its strain eigenvalues in increasing order are $c(t)(-p, 1, p-1)$, and its nonzero vorticity is aligned with the intermediate eigendirection. Its full characteristic polynomial is*

$$\det(rI - A(t)) = (r - c(t))^2 (r + 2c(t)),$$

whereas

$$\sigma(A(t)) = c(t), \qquad \rho(A(t)) = \frac{1}{\sqrt{p^2 + 1 + (p-1)^2}}. \tag{4}$$

For every $\epsilon > 0$ a choice of $p > 2$ makes $0 < \rho(A(t)) < \epsilon$ for all $0 \le t < 1$, while $\sigma(A(t)) \to \infty$ as $t \uparrow 1$. No positive uniform lower threshold for this normalized quantity holds on these exact profiles.

Proof. The middle coordinate of M is one. On the first and third coordinates, direct multiplication and $a^2 = p^2 - p - 2$ give

$$M^2 + M = 2I, \qquad \operatorname{tr} M = 0, \qquad \operatorname{tr} M^2 = 6.$$

Thus $-M^2 + \operatorname{tr}(M^2) I/3 = M$. Since $c' = c^2$ and the ODE right side is homogeneous of degree two, $A' = c^2 M$ is precisely that right side. Also

$$S(M) = \operatorname{diag}(-p, 1, p-1), \qquad \omega(M) = (0, 2a, 0).$$

Multiplying by $c > 0$ proves the strain ordering and alignment and gives $|\omega(A)|^2 = 4a^2 c^2$ and $S(A)\omega(A) = c\omega(A)$. The stretching formulas (4) follow, since $\|S(A)\|_F^2 = c^2(p^2 + 1 + (p-1)^2)$.

The determinant factors as

$$\det(rI - M) = (r-1)((r+p)(r-p+1) + a^2) = (r-1)^2 (r+2).$$

Rescaling gives the stated polynomial for A. The matrix is nonnormal: the $(1,3)$ entries of $M^{\mathsf{T}} M$ and $M M^{\mathsf{T}}$ are $-a(2p-1)$ and $a(2p-1)$, respectively, and are unequal. Hence the varying symmetric spectrum does not contradict the fixed full spectrum.

For $\epsilon > 0$, choose $p = 2 + \epsilon^{-1}$ and the specified a. The denominator in (4) is strictly larger than p, and $\epsilon p = 2\epsilon + 1 > 1$, so $0 < \rho < 1/p < \epsilon$. Meanwhile $c(t) \to \infty$ as $t \uparrow 1$, proving raw stretching divergence along each selected trajectory. □

For example $p = 3, a = 2$ gives strain ratios $-3 : 1 : 2$; $p = 4, a = \sqrt{10}$ gives $-4 : 1 : 3$ and normalized stretching $1/\sqrt{26}$. A particular strain ratio is therefore not universal even within this elementary self-similar family. Cantwell's general restricted-Euler solution already distinguishes full-gradient and strain/rotation geometry [1]. The examples here are exact calibration data, not a new solution theory.

3 An affine realization and its spatial limitation

Proposition 3.1. *With the same parameters, define on $\mathbb{R}^3$, for $t < 1$,*

$$u(t,x) = c(t)M_{p,a}x, \qquad p_{\mathrm{aff}}(t,x) = -c(t)^2|x|^2.$$

These smooth fields satisfy

$$\nabla \cdot u = 0, \qquad \partial_t u + (u\cdot\nabla)u + \nabla p_{\mathrm{aff}} = \nu\Delta u$$

for every fixed $\nu \geq 0$, pointwise on this time interval. They are not periodic and $u(t,\cdot) \notin L^2(\mathbb{R}^3)$.

Proof. The spatial gradient is cM, the divergence is $c\,\mathrm{tr}\,M = 0$, and $\Delta u = 0$. Moreover

$$\partial_t u = c^2 Mx, \qquad (u\cdot\nabla)u = c^2M^2x, \qquad \nabla p_{\mathrm{aff}} = -2c^2x.$$

The momentum identity is therefore $c^2(M + M^2 - 2I)x = 0$. Its pressure Hessian is $P = -2c^2I$ and satisfies $-\mathrm{tr}\,P = 6c^2 = \mathrm{tr}(A^2)$, so the Poisson convention is consistent. The middle velocity coordinate is $u_2 = cx_2$, which is not periodic. Since $c > 0$, integrating $|u|^2 \geq c^2x_2^2$ over expanding boxes gives an infinite integral. Thus these fields do not meet the whole-space finite-energy hypothesis. □

They also provide a local pressure-sign test. Define $q = \omega^{\mathsf{T}}S\omega$, $b = |S\omega|^2$ and $h = \omega^{\mathsf{T}}P\omega$ for this affine field. Directly,

$$q = 4a^2c^3, \qquad b = 4a^2c^4, \qquad h = -8a^2c^4.$$

Since q is spatially constant and $c' = c^2$,

$$(\partial_t + u\cdot\nabla)q = 12a^2c^4 = b - h \neq b + h.$$

This equality is a computation for these fields, not a proposed general formula with all other terms suppressed. It calibrates the sign of the displayed pressure contraction locally; it supplies no finite global integrals of the affine field.

4 A trace obstruction to mixed pressure conventions

The strain right side to be tested is the explicit matrix expression

$$R_+(A,P) = -S^2 - W^2 - P + \frac{\mathrm{tr}(S^2) + |\omega|^2/2}{3}I. \tag{5}$$

It is this expression, including the isotropic plus sign and the meaning of P, that the next result excludes as a trace-preserving incompressible strain evolution.

Theorem 4.1. *For every real A, P satisfying* (1)*,*

$$\mathrm{tr}\,R_+(A,P) = \|A\|_F^2. \tag{6}$$

In particular it is strictly positive if $A \neq 0$. Adding a trace-free matrix correction cannot restore trace zero. The full-pressure expression and the trace-free-pressure expression

$$R_{\mathrm{full}}(A,P) = -S^2 - W^2 - P, \tag{7}$$

$$R_{\mathrm{dev}}(A,P_0) = -S^2 - W^2 - P_0 + \frac{\mathrm{tr}(S^2) - |\omega|^2/2}{3}I \tag{8}$$

are trace-consistent under their respective conventions. Under (1)*,* (7) *and* (8) *agree.*

Proof. The symmetric/skew splitting is Frobenius orthogonal. The skew matrix associated with the above vorticity has squared Frobenius norm $|\omega|^2/2$, and hence

$$\begin{aligned}\operatorname{tr} W^2 &= -|\omega|^2/2,\\ \operatorname{tr} A^2 &= \operatorname{tr} S^2 - |\omega|^2/2,\\ \|A\|_F^2 &= \operatorname{tr} S^2 + |\omega|^2/2 = \operatorname{tr} A^2 + |\omega|^2.\end{aligned}$$

For example $|\omega|^2$ is the sum over $i<j$ of $(A_{ij}-A_{ji})^2$, while the off-diagonal contribution to $\operatorname{tr} A^2$ is $2\sum_{i<j} A_{ij}A_{ji}$; adding proves the last identity directly. Taking the trace of (5) gives $|\omega|^2 - \operatorname{tr} P$. Substituting (1) proves (6). The Frobenius squared norm is positive unless all entries vanish, and a trace-free addition leaves this trace unchanged.

The trace of (7) is $-\operatorname{tr} A^2 - \operatorname{tr} P$, which is zero under (1). The trace of (8) is zero because $\operatorname{tr} P_0 = 0$ and $\operatorname{tr}(S^2) - |\omega|^2/2 = \operatorname{tr} A^2$. Finally $P = P_0 + \operatorname{tr}(P)I/3 = P_0 - \operatorname{tr}(A^2)I/3$ under the constraint; inserting this in (7) gives (8). □

If $\operatorname{tr} A = 0$, the strain trace must remain zero along an incompressible evolution. Theorem 4.1 thus gives an immediate nonzero-state obstruction to (5). The theorem is a consistency test, not a derivation or closure of a nonlocal pressure law.

5 A periodic positive-viscosity pressure-sign test

On the 2π-periodic three-torus let

$$U(x,y,z) = (\sin z + \cos y,\ \sin x + \cos z,\ \sin y + \cos x).$$

This is the classical unit-parameter ABC profile; the same convention is used in [3]. Write $\langle f\rangle = (2\pi)^{-3}\int_{[0,2\pi]^3} f$ for normalized spatial average. For a smooth velocity v with strain S_v and vorticity ω_v, define

$$Q(v) = \langle \omega_v^{\mathsf{T}} S_v \omega_v\rangle, \quad B(v) = \langle |S_v\omega_v|^2\rangle, \quad H(v,p) = \langle \omega_v^{\mathsf{T}}(\nabla^2 p)\omega_v\rangle,$$

and the specific viscous contraction

$$J(v) = \left\langle \sum_{k,i,j} (S_v)_{ij}(\partial_k\omega_{v,i})(\partial_k\omega_{v,j})\right\rangle.$$

These are ordinary derivatives and integrals of smooth periodic fields.

Theorem 5.1. *For each $\nu>0$, the fields*

$$u(t) = e^{-\nu t}U, \qquad p(t) = -\tfrac{1}{2}|u(t)|^2$$

are a smooth periodic Navier–Stokes solution for all $t\ge 0$. They satisfy $Q(u(t))=0$ for every t, whereas at $t=0$,

$$B(U) = \tfrac{3}{2}, \qquad H(U,-|U|^2/2) = \tfrac{3}{2}, \qquad J(U) = 0. \tag{9}$$

In particular the identity

$$\frac{d}{dt}Q(u(t)) = B(u(t)) + H(u(t),p(t)) - 2\nu J(u(t)) \tag{10}$$

fails at $t=0$: its sides are zero and three respectively. Reversing the pressure sign makes the two sides agree on this test at $t=0$, without asserting a general evolution formula from this agreement.

Proof. Differentiation gives $\nabla \cdot U = 0$, $\nabla \times U = U$ and $\Delta U = -U$. For a Beltrami field, $(U \cdot \nabla)U = \nabla K$, $K = |U|^2/2$, by the vector identity $(U \cdot \nabla)U = \nabla K - U \times (\nabla \times U)$. Moreover $S_U U = \nabla K$, because the skew part of ∇U annihilates its vorticity U. For $u = e^{-\nu t}U$, the nonlinear advection is canceled by ∇p, and $\partial_t u = -\nu u = \nu \Delta u$. This proves the PDE assertion.

Periodic integration by parts and $\nabla \cdot U = 0$ give $Q(U) = \langle U \cdot \nabla K \rangle = 0$. At time t, stretching is scaled by $e^{-3\nu t}$, so $Q(u(t)) = 0$ and its derivative is zero. The kinetic gradient is

$$\nabla K = (\cos x \cos z - \sin x \sin y,\ \cos y \cos x - \sin y \sin z,\ \cos z \cos y - \sin z \sin x).$$

Each component has mean square $1/4+1/4 = 1/2$; the cross terms have zero average by separation of variables. Since $S_U U = \nabla K$, this proves $B(U) = 3/2$. For the pressure contraction, integrating $U_i U_j \partial_i \partial_j (-K)$ by parts in i gives

$$H(U, -K) = -\langle ((U \cdot \nabla)U) \cdot \nabla(-K) \rangle = \langle |\nabla K|^2 \rangle = \tfrac{3}{2}.$$

All boundary terms cancel by periodicity.

It remains to check the precise viscous contraction, rather than omit it. The strain has zero diagonal and

$$S_{12} = \tfrac{1}{2}(\cos x - \sin y), \quad S_{13} = \tfrac{1}{2}(\cos z - \sin x), \quad S_{23} = \tfrac{1}{2}(\cos y - \sin z).$$

As $\omega_U = U$, the three terms $(\partial_k U)^{\mathsf{T}} S_U (\partial_k U)$, for $k = x, y, z$, are

$$\begin{aligned} &-(\cos y - \sin z)\cos x \sin x, \\ &-(\cos z - \sin x)\sin y \cos y, \\ &-(\cos x - \sin y)\cos z \sin z. \end{aligned}$$

Each has zero normalized average; for example integration in x kills the first. Thus $J(U) = 0$. These values prove both the failure of (10) and the limited sign-reversal calibration. □

The affine sign test and periodic test address different possible loopholes. The former is local and has no finite whole-space energy; the latter is smooth, periodic, and has arbitrary positive viscosity. Neither asserts that all pressure contractions have one sign, or that changing one sign establishes a complete general strain balance.

6 Formal correspondence and contribution status

Every formal result is frozen at commit `bab688bb6140cd6316c80680095a914ea0d346db`. The family and its small-normalized-stretching quantifiers are in `RestrictedEulerAlignedGrowth`, `RestrictedEulerSpectralDistinction` and `RestrictedEulerArbitraryDepletion`. Actual affine derivatives, momentum and pressure appear in `AffineNavierStokesProfiles` and `AffineStretchingPressureBalance`. The trace test is in `PressureStrainTraceObstruction`. The real periodic derivatives, means and viscous contraction are assembled in `ABCNavierStokesFlow`, `RealBeltramiMeans`, `BeltramiViscousTrace` and `PeriodicPressureSignCounterexample`. Exact declaration locations, hashes and human proof reorganizations are in the accompanying theorem map. In particular the direct trigonometric proof of $J(U) = 0$ replaces the formal invariant-trace transport proof. Infinite affine energy is the elementary deduction displayed above; the mapped source explicitly proves spatial unboundedness and nonperiodicity, not a named integral theorem.

These examples belong to classical restricted-Euler and Beltrami theory. Cantwell's primary paper was inspected for its model, solution scope and distinction between full-gradient and strain geometry. The primary 2024 pressure-Hessian paper fixes the full/deviatoric conventions; no conflict with its stated equations is alleged. The ABC profile attribution was checked against the primary Xin–Yu–Zlatoš model statement. The examples test the explicit formulas written here, rather than an unspecified published theorem. Their value is reproducible diagnostic mathematics and a formalization companion; novelty or priority for any formulation remains unresolved. No Navier–Stokes regularity, finite-energy blow-up or Millennium equivalence claim follows. Complete baseline verification evidence was hash-checked without a redundant full Lean build.

References

[1] B. J. Cantwell, *Exact solution of a restricted Euler equation for the velocity gradient tensor*, Phys. Fluids A 4 (1992), 782–793. doi:10.1063/1.858295.

[2] P.-F. Yang, H. Xu, A. Pumir and G. W. He, *Structure and role of the pressure Hessian in regions of strong vorticity in turbulence*, J. Fluid Mech. 983 (2024), R2. doi:10.1017/jfm.2024.143.

[3] J. Xin, Y. Yu and A. Zlatoš, *Periodic orbits of the ABC flow with* $A = B = C = 1$, arXiv:1601.02724 (2016). Profile comparison only; no orbit result is used here.